

ETHAN JENSEN

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Portfolio: ethanjensn.github.io/portfolio/

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, TypeScript, SQL

Frameworks & Technologies: React, REST APIs, WebSockets, Docker, PostgreSQL, Prisma, Git, FFmpeg, GStreamer

AI/ML & Tools: YOLO, Reinforcement Learning, CNNs, NVIDIA DeepStream, Isaac Lab

PROFESSIONAL EXPERIENCE

AI/ML Software Engineering Intern | SoundSafe.ai

Jun 2025 – May 2026 | Galena, IL

- Progressively assumed end-to-end software engineering responsibilities across AI/ML systems, backend services, and production deployments.
- Architected and deployed a real-time multi-stream violence detection system using YOLO-based vision models and event-driven backend services.
- Designed low-latency video ingestion and inference pipelines using FFmpeg, GPU acceleration, and GStreamer.
- Led end-to-end system development across model training, backend processing, and frontend monitoring dashboards.
- Designed low-latency multi-drone detection pipelines using RF-DETR with a DINOv2 backbone for real-time inference.
- Built a real-time action classification pipeline combining YOLO pose estimation with a Transformer-based temporal model.
- Developed and optimized a Reinforcement Learning (RL) agent in NVIDIA Isaac Lab for LiDAR-based autonomous navigation, improving simulation efficiency through Direct Tensor Access.
- Engineered a custom CNN-based OCR system using EMNIST datasets for live video streams.

Commission Sales Associate | Slumberland Furniture

Dec 2022 – Sept 2025 | Dubuque, IA

- Managed high-volume customer relationships to consistently exceed performance and sales quotas.

Repair Technician | CPR Cell Phone Repair

July 2021 – Dec 2022 | Dubuque, IA

- Diagnosed hardware and software issues for mobile devices, consoles, and computers while managing inventory and technical operations.

Warehouse Associate | Slumberland Furniture

Oct 2020 – July 2021 | Dubuque, IA

- Directed warehouse logistics and assembly while supervising team efficiency and inventory accuracy.

FEATURED PROJECTS | ethanjensn.github.io/portfolio/

Full-Stack RSVP Management Platform | React, PostgreSQL, Prisma

- Developed a production-ready RSVP platform with secure API routes, persistent guest management, and mobile-responsive UI.
- Built backend services using PostgreSQL, Prisma, and server-side validation with an admin analytics dashboard.

Autonomous Navigation (RL) | NVIDIA Isaac Lab

- Trained a 4-wheel vehicle using PPO to navigate waypoints while avoiding obstacles via LiDAR-based ray casting.

Violence Detection Pipeline | YOLO, CNN + Transformer

- Built a real-time system using YOLO pose estimation and a Transformer model to classify violent vs non-violent actions.

Automatic License Plate Recognition | YOLO, PaddleOCR

- End-to-end system for detecting and reading plates from video streams, optimized for scalability and storage.

Real-Time Alphanumeric OCR | CNN

- Lightweight dual-model CNN for real-time camera character recognition with hot-swappable model capabilities.

EDUCATION

Bachelor of Science in Computer Science

Southern New Hampshire University — Expected May 2027